

Part C: Running with Docker

Create a Dockerfile:

```
$nano Dockerfile
FROM gcc:latest
COPY hello-d.c /app/hello-d.c
RUN gcc /app/hello-d.c -o /app/hello
CMD ["/app/hello"]
```

Then build and run:

```
$ sudo docker build -t hello-container .
$ sudo /usr/bin/time -f "Execution Time (Docker): %e seconds"
docker run --rm hello-container
```

This produces the expected greeting and reports execution time.

Part D: Running with Wasm

Compile using the WASI SDK:

```
$WASI_SDK_PATH/bin/clang --target=wasm32-wasi -O2 -o hello.
wasm hello-w.c
```

Run using Wasmtime:

```
/usr/bin/time -f "Execution Time (WASM): %e seconds" wasmtime
hello.wasm
```

Part E: Comparative analysis

Refer to the following table and Figures 2 and 3 for the comparative analysis.

Metric	Docker container	WebAssembly (Wasm)
Program output	Hello from Docker Container!	Hello from Web Assembly!
Execution time	0.84 seconds	0.01 seconds
Binary/Image size	1.45GB (Docker image)	4.4KB (<i>hello.wasm</i> file)
Startup overhead	High (container startup, GCC)	Very low (Wasm runtime only)
Command used	<i>docker run --rm hello-container</i>	<i>wasmtime hello.wasm</i>

This hands-on comparison between Docker containers and WebAssembly (Wasm) illustrates the trade-offs and strengths of each technology. By using the same simple C program compiled and executed in both environments, we can assess measurable metrics such as execution time, binary size, and runtime behaviour.

The Docker-based build, despite being straightforward to set up with a Dockerfile using the gcc base image, resulted in a significantly larger image (1.45GB) and a noticeably slower execution time (~0.84 seconds) as shown in Figure 2. This overhead stems primarily from the inclusion of a full Linux base image and the GCC toolchain — even for a lightweight program.

```
osfy@Shaanu:~/docker-vs-wasm$ sudo docker build -t hello-container .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 79.58MB
Step 1/4 : FROM gcc:latest
----> 50fd257342f0
Step 2/4 : COPY hello-d.c /app/hello-d.c
----> Using cache
----> 30945ef0211f
Step 3/4 : RUN gcc /app/hello-d.c -o /app/hello
----> Using cache
----> 34aae7932d1e
Step 4/4 : CMD ["/app/hello"]
----> Using cache
----> a66b2a5bbd8e
Successfully built a66b2a5bbd8e
Successfully tagged hello-container:latest
osfy@Shaanu:~/docker-vs-wasm$ sudo docker images hello-container
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
hello-container      latest          a66b2a5bbd8e   12 minutes ago 1.45GB
osfy@Shaanu:~/docker-vs-wasm$ sudo /usr/bin/time -f "Execution Time (Docker): %e seconds" docker run --rm hello-container
Hello from Docker Container!
Execution Time (Docker): 0.84 seconds
```

Figure 2: Docker container preparation and execution